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Projection Mapping System and Apparatus

Abstract

A marking and display system includes a scanning apparatus, comprising a processor in data communication with a networking device, at least one input/output device, and computer memory, the computer memory comprising a program having machine readable instructions that, when effected by processor, perform the following steps: (a) scan an area of interest for a first scan; (b) determine the presence of at least one abnormality at the area of interest; and (c) provide a marking at an area substantially adjacent the at least one abnormality.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/394,229, filed Dec. 22, 2023, which will issue as U.S. Pat. No. 12,317,007, which is a divisional of U.S. patent application Ser. No. 16/701,727, filed Dec. 3, 2019, which issued as U.S. Pat. No. 11,856,336, and claims priority to U.S. Provisional Patent Application No. 62/782,311, filed Dec. 19, 2018. The '727 application is also a continuation-in-part of U.S. patent application Ser. No. 16/278,628, filed Feb. 18, 2019, which issued as U.S. Pat. No. 10,495,961 and which is a continuation-in-part of U.S. patent application Ser. No. 16/036,772, filed Jul. 16, 2018, which issued as U.S. Pat. No. 10,212,404 and which is a continuation-in-part of U.S. patent application Ser. No. 15/622,959, filed Jun. 14, 2017, which issued as U.S. Pat. No. 10,027,937. Each of the foregoing applications is incorporated herein by reference in its entirety.

FIELD OF INVENTION

[0002] The invention relates to projection mapping apparatus. More specifically, the invention relates to an apparatus for projecting an image onto a surface. The image may or may not be human readable.

BACKGROUND

[0003] Projection displays have been around for several years in many different forms. While many industries take advantage of projection technology, one industry where such technology has been largely ignored is hand tools. Generally speaking, hand tools have seen relatively few advancements over the years. This is especially true when it comes to tape measures.

[0004] Currently, tape measures are effective for their intended purpose. However, they tend to be bulky and somewhat difficult to use, as a user must both lay out the tape measure upon a surface and mark the surface while attempting to hold the tape measure in position. This often results in frustration, especially when the tape measure becomes dislodged from its desired position, twists, or the user has to take measure many different surfaces. Accordingly, it would be desirable to have a projection tape measure which allows a user to measure a surface without requiring him or her to physically hold any device.

SUMMARY

[0005] The following presents a simplified summary of the invention in order to provide a basic understanding of some aspects of the invention. This summary is not an extensive overview of the invention. It is not intended to identify critical elements of the invention or to delineate the scope of the invention. Its sole purpose is to present some concepts of the invention in a simplified form as a prelude to the more detailed description that is presented elsewhere herein.

[0006] In one embodiment, a projection system includes a projection apparatus embodied in a housing secured to a user. The projection apparatus has a processor in data communication with a networking device, at least one input/output device, and computer memory. The computer memory includes a program with machine readable instructions that, when effected by processor, perform the following steps: (a) determine an edge of a surface to be measured; (b) project an initial image onto the surface, the initial image being based on a predetermined set of conditions; (c) determine a substantially perpendicular distance D1 between the projection apparatus and the surface; (d) determine a distance D2 between the projection apparatus and the edge of the surface; (e) calibrate the initial image based on the distances D1 and D2 determined in steps (c) and (d); and (f) project an updated image onto the surface.

[0007] In another embodiment, a projection system comprises a reference device comprising a

processor in data communication with a networking device and at least one input/output device. The reference device is placed at an edge of a surface to be measured. The system further includes a projection apparatus embodied in a housing secured to a user. The projection apparatus comprises a processor in data communication with a networking device, at least one input/output device, and computer memory. The computer memory includes a program having machine readable instructions that, when effected by processor, performs the following steps: (a) projecting an initial image onto the surface, the initial image being based on a predetermined set of conditions; (b) determining a substantially perpendicular distance D1 between the projection apparatus and the surface; (c) determining a distance D2 between the projection apparatus and the reference device; (d) calibrating the initial image based on the distances D1 and D2 determined in steps (b) and (c); (e) projecting an updated image onto the surface; and (f) repeating steps (b)-(e). The reference device and the projection apparatus communicate over a network.

[0008] In still another embodiment, a projection system has a reference device and a projection apparatus embodied in a housing secured to a user. The projection apparatus includes a processor in data communication with a networking device, at least one input/output device, and computer memory. The computer memory includes a program having machine readable instructions that, when effected by processor, perform the following steps: (a) projecting an initial image onto the surface, the initial image being based on a predetermined set of conditions; (b) determining a substantially perpendicular distance D1 between the projection apparatus and the surface; (c) determining a distance D2 between the projection apparatus and the reference device; (d) calibrating the initial image based on the distances D1 and D2 determined in steps (b) and (c); (e) projecting an updated image onto the surface; and (f) repeating steps (b)-(e).

[0009] In still another embodiment, a marking and display system includes a first array display apparatus having a viewing angle. The display apparatus has a processor in data communication with a networking device, at least one input/output device, and computer memory. The computer memory includes a program having machine readable instructions that, when effected by processor, iteratively perform the following steps: (a) determining the presence of a distant surface; (b) marking, within the viewing angle, an edge of the distant surface and a plurality of locations on the distant surface; (c) displaying an initial array onto an array receiving panel having a first panel edge and a second panel edge; (d) determining a distance D1.sub.a-D1.sub.n between the display apparatus and each of the plurality of locations on the distant surface; (e) determining a forward distance D2 between the display apparatus and the array receiving panel; (f) determining a distance D3 between the display apparatus and the edge of the distant surface; (g) determining a distance D4 between the first panel edge and the second panel edge; (h) calibrating the initial array on the array receiving panel based on the distances D1.sub.a-D1.sub.n, D2, D3, and D4; and (i) projecting an updated array onto the array receiving panel.

[0010] In a further apparatus, a marking and display system has an array display apparatus with a viewing angle. The system comprises a processor in data communication with a networking device, at least one input/output device, and computer memory, the computer memory comprising a program having machine readable instructions that, when effected by processor, iteratively perform the following steps: (a) determining the presence of a distant surface; (b) marking, within the viewing angle, an edge of the distant surface and a plurality of locations on the distant surface; (c) displaying an initial array onto an array receiving panel having a first panel edge and a second panel edge; (d) determining a distance D1.sub.a-D1.sub.n between the display apparatus and each of the plurality of locations on the distant surface; (e) determining a forward distance D2 between the display apparatus and the array receiving panel; (f) determining a distance D3 between the display apparatus and the edge of the distant surface; (g) determining a distance D4 between the first panel edge and the second panel edge; (h) calibrating the initial array on the array receiving panel based on the distances D1.sub.a-D1.sub.n, D2, D3, and D4; (i) projecting an updated array onto the array receiving panel; (k) repeating steps (a) through (f) and (h). At step (c), the initial

array displayed on the array receiving panel is substituted with the updated array from step (j); and the updated array at step (j) is replaced by a second updated array.

[0011] In still yet another embodiment, a marking and display system includes an array display apparatus having a viewing angle. The system further comprises a processor in data communication with a networking device, at least one input/output device, and computer memory, the computer memory comprising a program having machine readable instructions that, when effected by processor, iteratively perform the following steps: (a) determining the presence of a distant surface; (b) marking, within the viewing angle, an edge of the distant surface and a location on the distant surface; (c) determining the presence of an environmental object of the distant surface; (d) displaying an initial array onto a windshield having a first edge and a second edge, the initial array being based on the environmental object; (e) determining a distance $D1$ between the display apparatus and the location on the distant surface; (f) determining a forward distance $D2$ between the display apparatus and the windshield; (g) determining a distance $D3$ between the display apparatus and the edge of the distant surface; (h) determining a distance $D4$ between the windshield first edge and second edge; (i) calibrating the initial array on the windshield based on the distances $D1$, $D2$, $D3$, and $D4$; and (j) projecting an updated array onto the array receiving panel.

[0012] According to still another embodiment, a light emitting system comprises a light emission apparatus spatially separated from a receiving panel. The light emission apparatus includes a processor in data communication with a first light source and computer memory, and the computer memory has machine readable instructions that, when effected by the processor, perform the following steps: (a) activate the first light source to emit a first light wave in the direction of the receiving plane at an emission angle $\Omega_{\text{sub.1}}$; (b) determine a distance $D1$ between the receiving plane and the light emission apparatus; (c) determine a first edge of the receiving plane; (d) determine a first receiving location of the first light wave on the receiving plane; (e) determine a distance $D2$ between the first edge of the receiving plane and the first receiving location; (f) adjust the first light source based on the distances $D1$, $D2$, and the emission angle $\Omega_{\text{sub.1}}$; and (g) activate the first light source to emit an adjusted first light wave in the direction of the receiving panel.

[0013] According to still yet another embodiment, a light emitting system has a light emission apparatus spatially separated from a receiving panel. The light emission apparatus includes a processor in data communication with a camera, a first light source, and computer memory, and the computer memory has machine readable instructions that, when effected by the processor, perform the following steps: (a) activate the camera to record the receiving panel; (b) activate the first light source to emit a first light wave in the direction of the receiving plane at an emission angle $\Omega_{\text{sub.1}}$; (c) determine a distance $D1$ between the receiving plane and the light emission apparatus; (d) determine a first edge of the receiving panel; (e) determine a first receiving location of the first light wave on the receiving panel; (f) determine a distance $D2$ between the first edge of the receiving panel and the first receiving location; (g) adjust the first light source based on the distances $D1$, $D2$, and the emission angle $\Omega_{\text{sub.1}}$; (h) activate the first light source to emit an adjusted first light wave in the direction of the receiving panel; and (i) transmit the recording from the camera to a sharing apparatus.

[0014] In a further embodiment, a light emitting system includes a light emission apparatus spatially separated from a receiving panel. The light emission apparatus has a processor in data communication with a first light source, a second light source, and computer memory. The computer memory includes machine readable instructions that, when effected by the processor, perform the following steps: (a) activate the first light source to emit a first light wave in the direction of the receiving plane at an emission angle $\Omega_{\text{sub.1}}$; (b) activate the second light source to emit a second light wave in the direction of the receiving plane at an emission angle $\Omega_{\text{sub.2}}$; (c) determine a distance $D1$ between the receiving plane and the light emission apparatus; (d) determine a first edge and a second edge of the receiving plane, the first and second edges being

parallel; (e) determine a first receiving location of the first light wave on the receiving panel; (f) determine a second receiving location of the second light wave on the receiving panel; (g) determine a preferred receiving location at the receiving panel; (h) adjust the first light source to emit an adjusted first light wave in the direction of the preferred receiving location; and (i) adjust the second light source to emit an adjusted second light wave in the direction of the preferred receiving location.

[0015] According to still yet another embodiment, a marking and display system includes a scanning apparatus, comprising a processor in data communication with a networking device, at least one input/output device, and computer memory, the computer memory comprising a program having machine readable instructions that, when effected by processor, perform the following steps: (a) scan an area of interest for a first scan; (b) determine the presence of at least one abnormality at the area of interest; and (c) provide a marking at an area substantially adjacent the at least one abnormality.

[0016] According to a further embodiment, a marking and display system includes a marking and scanning apparatus comprising a processor in data communication with a networking device, at least one input/output device, and computer memory. The computer memory includes a program having machine readable instructions that, when effected by processor, perform the following steps: (a) project a first image onto an area of interest; (b) scan the first image at the area of interest; (c) determine the presence of at least one abnormality at the area of interest; and (d) project a second image substantially adjacent the at least one abnormality. The first image is based on a predetermined set of conditions. And the second image is calibrated based on a determination of an edge of the abnormality.

[0017] According to yet another embodiment, a marking and display system includes a projection apparatus embodied in a housing. The projection apparatus includes a processor in data communication with computer memory. The computer memory has a program having machine readable instructions that, when effectuated by the processor, perform the following steps: (a) receive information regarding an object on a scanning area from a user; (b) project a first image onto the scanning area; (c) scan the scanning area to determine at least one characteristic about the object on the scanning area; and (d) project a second image onto the scanning area, wherein the second image is based on the first image and the information provided by the user.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 is a schematic illustration of a projection apparatus and system according to an embodiment of the invention.

[0019] FIG. 2A is a front view of an embodiment of the projection apparatus of FIG. 1.

[0020] FIG. 2B is a side view of the projection apparatus of FIG. 2A.

[0021] FIG. 3 is a perspective view of the projection apparatus of FIG. 2A incorporated into a hat.

[0022] FIG. 4A is a side view of another embodiment of the projection apparatus of FIG. 1.

[0023] FIG. 4B is a front view of the projection apparatus of FIG. 4A.

[0024] FIG. 5 is a perspective view of the projection apparatus of FIG. 4A incorporated onto a pair of glasses.

[0025] FIG. 6 is a flowchart illustrating various steps performed by projection apparatus systems according to an embodiment of the invention.

[0026] FIG. 7 is a perspective view of a projection apparatus system according to an embodiment of the invention.

[0027] FIG. 8 is a perspective view of a projection apparatus system according to another embodiment of the invention.

[0028] FIG. **9** is a perspective view of a projection apparatus system according to still another embodiment of the invention.

[0029] FIG. **10** is a perspective view of a projection apparatus system according to another embodiment of the invention.

[0030] FIG. **11** is a flowchart illustrating various steps performed by projection apparatus systems according to an embodiment of the invention.

[0031] FIG. **12** is a schematic illustration of a light emitting apparatus and system according to an embodiment of the invention.

[0032] FIG. **13A** is a top view of a light emitting apparatus according to an embodiment of the invention.

[0033] FIG. **13B** is a side view of the light emitting apparatus of FIG. **13A**.

[0034] FIGS. **14A-B** are respective top views of a light emitting apparatus showing convergence of light waves at a focal point.

[0035] FIG. **15** is a flowchart illustrating various steps performed by a light emitting apparatus according to an embodiment of the invention.

[0036] FIG. **16** is a flowchart illustrating various steps performed by a light emitting apparatus according to another embodiment of the invention.

[0037] FIG. **17** is a perspective view of a projection apparatus system according to still another embodiment of the invention.

DETAILED DESCRIPTION

[0038] Currently, the majority of hand tools do not incorporate any type of projection system. One exception is distance finders which use ultrasonic and/or laser reflection techniques to determine a distance from the user to a surface. The user points the tool at a surface, presses a button to activate the laser, and the tool measures the distance to the location where the laser is pointed. While these devices are useful for determining the distance to a point, they are not tape measures. A tape measure cannot be substituted for a laser pointer where the user desires to, for example, mark a surface for cutting.

[0039] Disclosed herein are embodiments of projection mapping apparatus which may be useful as a tape measure projection device. Those of skill in the art shall understand that while reference is made herein to apparatus that project tape measures, other projection apparatus are contemplated within the scope of the invention and will become apparent from the description provided herein.

[0040] In one embodiment, a projection mapping system and apparatus includes a projection apparatus **200** which may be configured to attach to a user's person or incorporated into an article worn by the user as described herein. Electronic components of the projection apparatus **200** are illustrated in FIG. **1**. The projection apparatus **200** includes a processor **205** communicatively coupled to a networking device **210**, one or more input/output devices **215**, and computer memory **220**. The processor **205** may be configured through particularly configured hardware, such as an application specific integrated circuit (ASIC), field-programmable gate array (FPGA), etc., and/or through execution of software (e.g., program **225**) to perform functions in accordance with the disclosure herein.

[0041] Memory **220** represents one or more of volatile memory (e.g., RAM) or non-volatile memory (e.g., ROM, FLASH, magnetic media, optical media, etc.). Although shown within the projection apparatus **200**, the memory **220** may be, at least in part, implemented as network storage external to the projection apparatus **200** which may be accessed via the network device **210**. The network device **210** may be implemented as one or both of a wired network interface and a wireless network interface (e.g., Wi-Fi, Internet, Bluetooth, etc.), as is known in the art.

[0042] The network device **210** may allow the projection apparatus **200** to communicate over a network **250** with a reference device **270**. The network **250** may be a wireless network, such as Wi-Fi, Bluetooth, or other wireless (or wired) network.

[0043] Program **225** may be stored in a transitory or non-transitory portion of the memory **220**.

Program **225** includes machine readable instructions that, when executed by the processor **205**, perform one or more of the functions of the device **200**. In embodiments, the program **225** may include instructions for calculating distances and scales and projecting an image onto a surface (e.g., surface **100**) as described in greater detail below with reference to FIG. 6.

[0044] An image database **223** may additionally be stored in the memory **220**, or alternately may be stored in remote storage accessible over the network **250**. The image database **223** may contain various images of tape measures which may be projected onto a surface **100** via the projection apparatus **200** according to the description provided herein. For example, there may be tape measure images that display measurements according to the metric system, and other tape measure images that display measurements according to the imperial system. Additionally, the user may prefer one color over another, and therefore there may be multiple images of tape measures in varying colors. Traditionalists may prefer the projection to show the tape measure in yellow, which others may prefer a different color (e.g., white, blue, green, orange, etc.). Accordingly, various images may be stored in the image database **223** accessible by the processor **205**.

[0045] The input/output device **215** may include one or more input and/or output devices which may be embodied in a single device or multiple devices. In one embodiment, the input/output device **215** includes at least a projector for projecting an image onto a surface (e.g., surface **100**). The input/output device **215** may additionally include a laser and/or a camera. Optionally, the input/output device **215** includes a UV laser for marking purposes, as is described below. In one embodiment, the input/output device **215** may further include a speaker. The input/output device **215** may still further include a button **215A** and **215A'** (FIGS. 2B and 4B), for example, to allow the user to interact with the apparatus **200** and/or the reference device **270**, as described herein.

[0046] The reference device **270** may have a processor **275** communicatively coupled to an input/output device **280** and a network device **285**. The network device **285** may allow the reference device **270** to communicate over the network **250** with the projection apparatus **200**.

[0047] The input/output device **280** may be an emitting device which emits a signal (e.g., over the network **250**) which may be received by the projection apparatus **200** in order to determine the distance between the reference device **270** and the projection apparatus **200**. Alternately, the reference device **270** may be in communication (e.g., wired or wirelessly) with the projection apparatus **200** to communicate a distance from the projection apparatus **200** to the reference device **270**.

[0048] The projection apparatus **200** may be embodied in a variety of different devices. In one embodiment, the projection apparatus **200** may be incorporated into a handheld device, similar to a laser measuring device or flashlight. In another embodiment, the projection apparatus **200** may be configured as a clip. FIGS. 2A, 2B, 4A, and 4B illustrate exemplary configurations of a clip. In FIG. 2A, the projection apparatus **200** is embodied in a clip having a housing **230** with apertures formed therein through which one or more input/output devices **215** may extend. It shall be understood by those of skill in the art that the input/output device(s) **215** may not extend all the way through the aperture, but the aperture may allow the input/output device(s) **215** to give and/or receive information to/from the system. The housing **230** further includes a button **215A** which the user may use to interact with the apparatus **200**. An arm **234** may rotatably attach to the housing **230** via a pin **232** (for example), and may be spring-loaded such that the arm **234** is maintained in a closed position unless the user forcibly opens the arm **234** (e.g., in order to attach the apparatus **200** to a wearable article). FIG. 3 shows a hard hat **400** with a projection apparatus **200** attached thereto. The projection apparatus **200** may alternately be attached to other wearable articles, such as a baseball cap or other type of hat, or to an article of clothing, such as to the user's collar.

[0049] In FIGS. 4A, 4B, and 5, an apparatus **200'** is configured for attachment to safety goggles **450** or glasses. The apparatus **200'** includes a housing **230'** and a clip **234'** which may be configured to engage with the nose bridge of the glasses **450**. The clip **234'** may be adjustable to ensure that the apparatus **200'** is securely attached to the goggles **450**. The apparatus **200'** includes

at least out input/output device **215**, and may additionally include a button **215A'** as shown in FIGS. **4B** and **5**.

[0050] The projection apparatus **200** may preferably be adjustable such that the input/output device **215** is appropriately oriented. For example, the housing **230** may be adjustably attached to a plate which may be secured to the arm **234** via the pin **232**. The position of the housing **230** may therefore be adjusted as necessary. Alternately, in embodiments, the projection apparatus **200** may be incorporated directly into items such as the hard hat or safety goggles.

[0051] The electronic components of the apparatus **200** may be battery operated, solar powered, or may receive power by any other means now known or later developed. In one embodiment, the apparatus **200** includes a rechargeable battery which may be recharged using solar power, electrical power, etc.

[0052] The features of the various components described herein shall be further understood by way of examples of the projection apparatus **200** in a use configuration. Referring now to FIG. **6**, an exemplary set of process steps is illustrated according to an embodiment of the invention. The process begins at step **600**, when the user determines that a surface requires a measurement. At step **602**, the user may activate the reference device **270** and places the reference device **270** such that it abuts one end of the surface **100**, as shown in FIG. **8**. Further discussion of the reference device **270** is provided below.

[0053] Moving on to step **604**, the projection apparatus **200** is activated. The projection apparatus **200** may be activated via, for example, a button **215A** on the apparatus **200**. The button **215A** may cause the apparatus **200** to turn to an "on" mode. Alternately, the apparatus **200** may be equipped with for example, a gyroscope which may detect movement of the user's head. When the user shakes his or her head, the apparatus **200** may be activated.

[0054] At step **606**, upon activation, the apparatus **200** may project an initial image **150** onto the surface **100** (see FIG. **7**). The initial image **150** may not yet be the user's desired depiction of a tape measure. For example, the projection apparatus **200** may be initially programmed to project an image from the image database **223** of a tape measure showing measurements according to the imperial measurement system. However, the user may prefer that the measurements be shown according to the metric system. Additionally, the initial image **150** may be programmed to project the initial image **150** based on pre-set reference conditions. However, the reference conditions may not accurately portray the position of the user in reference to the surface **100**, and therefore, the initial image **150** may project inaccurate units of measurement (e.g., one inch as projected in the initial image **150** is not a true representation of one inch).

[0055] Nevertheless, the process moves to step **608**, wherein the user selects the desired tape measure image from the image database **223**. The user may be able to change the image in order to customize the system of measurement (e.g., imperial or metric) by toggling through the images stored on the image database **223** to arrive at the desired image, e.g., via the button **215A** or **215A'**. Additionally, the user may be able to select an image that is color preferential to the user. Once the user arrives at his or her preferred image, the user's preferences may be automatically stored in the memory **220** and recalled each time the user activates the projection apparatus **200**.

[0056] Moving on to step **610**, once the user has selected his or her preferred image, the processor **205** begins the process of calibrating the projected image to the user's position in relation to the surface **200**. As noted above, the projection apparatus **200** may be programmed to project an initial image **150** based on pre-set reference conditions at which the projected image portrays units of measurement in accurate one-inch units. For example, referring again to FIG. **7**, the reference conditions may assume that a distance **D1** from the projection apparatus **200** to the surface **100** is equal to 2 feet, and the distance **D2** from the edge of the surface **105** to the center point **CP1** of the projected image is 6 inches. At these conditions, a reference angle α between the user and the end of the surface **105**, calculated using the equation $\cot \theta = D1/D2$, is 14.04 degrees. At these conditions, the projection apparatus **200** may project an image of a tape measure, wherein the

increments of measurement are shown at exactly 1 inch. However, it is unlikely that the user will maintain these reference conditions. As the user moves closer to or away from the surface **100**, the scale of the image must be altered such that the projected image still accurately displays the units of the tape measure. Accordingly, the position of the user relative to the surface **100** must be determined.

[0057] At step **610a**, the processor **205** causes the projection apparatus **200** to activate one or more output devices **215** (e.g., a laser) to measure the actual distance **11** (FIG. **8**) from the user U (having the projection apparatus **200** attached thereto) to the center point CP at surface **100**. Using techniques known to those of skill in the art, the distance **11** may be ascertained.

[0058] At step **610b**, the projection apparatus **200** communicates with the reference device **270**, e.g., over the network **250** via networking devices **210** and **285**, respectively, to determine the location of the reference device **270**. Knowing the location of the reference device **270**, the processor **205** is able to determine the distance **12** between the projection apparatus **200** and the reference device **270**.

[0059] The reference device **270** may, in one embodiment, be a smart phone equipped with downloadable software (e.g., a smart phone application) that allows it to communicate with the projection apparatus **200**. In order to use the smart phone as the reference device **270**, the user may access the smart phone application on the phone, which may, among other things, activate the networking device **285** such that the projection apparatus **200** may communicate with the phone. For example, after the distance **12** is determined, a planar verification display pattern may be projected onto the surface intended to be measured. The projected display pattern could show geometric shapes with known ratios such as squares, circles, etc. The projected sequence of predictable patterns may be shown sequentially along the surface in order to verify that the surface is flat (or being measured on a common plane). Adjustments in software can be made to adjust for planar integrity for any skewing of the plane of measurement. In one embodiment, the software may be configured to adjust for contoured surfaces utilizing 3D mapping and marking techniques including stereoscopic viewing of the projected display pattern.

[0060] In embodiments, the reference device **270** may be omitted. For example, in one embodiment, the projection apparatus **200** may include a camera (e.g., as an input/output device **215**). Using traditional gradient image processing techniques, the processor **205** may be able to ascertain the end **105** of the surface **100**, and thus determine the length **12**. In still another embodiment, the end **105** of the surface **100** may be marked with a marker (e.g., via a marking device such as a UV laser, etching, red dot marking, or via any other marker currently in use or later developed). The projection apparatus **200** may be configured to recognize the marker in order to ascertain the position of the end **105** of the surface **100**, and thus determine the length **12**.

[0061] Moving on, at step **610c**, the processor **205** determines the missing length **13** based on the Pythagorean Theorem for right triangles: $a^2 + b^2 = c^2$. The unit of the length **13** may be determined based on the user's selection of the system of measurement. For example, if the user selects the imperial system, then the lengths **11**, **12**, and **13** may be measured and determined in inches and feet. Alternately, if the user selects the metric system, then the lengths **11**, **12**, and **13** may be measured and determined in centimeters and meters. More advanced non-linear methods may alternately be used to calculate the measurement of length **13**. One such reiterative method involves differential calculus which would allow contoured measurements of surfaces that are not residing on a single flat plane.

[0062] Having determined the length **13**, the processor **205** then calculates the angle θ between the user and the end of the surface **105**. For example, if the user is 2.7' (32.4 inches) from the surface **100** (**11**), and the distance **12** is determined to be 40 inches, then the angle θ between the user and the end of the surface **105** as determined by trigonometric principles is 35.9 degrees. Once the processor **105** determines the length **13**, and the angle θ , the process moves to step **610d**. The processor **205** is not limited to a single angle θ measurement in order to provide more precise

results in measurement. Multiple calculations may be made in succession in order to provide a desired precision resulting measurement.

[0063] At step **610d**, the processor **205** determines the factor by which the size of the projected image must be altered such that the units of measurement of the projected image correspond with the length **13** determined at step **610c**. In our example, the ratio of the reference angle to the actual angle is 0.74. Therefore, the image must be scaled down in length by a factor of 0.39. Using methods of scaling known to those of skill in the art, the program may be configured to scale the projected image **150** by the appropriate factor.

[0064] At step **610e**, the projection apparatus **200** projects the altered image onto the surface **100**, wherein the altered image is appropriately scaled based on the position of the user to the surface **100** as described above.

[0065] In one embodiment, illustrated in FIG. **9**, the user may be a distance L from the end of the surface **105**. Those of skill in the art may recognize that in order for the projected image to be as accurate as possible, it may be desirable for the user to be positioned such that the projection apparatus **200** is substantially perpendicular to the surface **100**. Therefore, to measure distances that are farther away from the end of the surface **105** (e.g., distances greater than the beam angle γ , or the angle of the beam from the projection apparatus **200**), the user may have to move into the general vicinity of the final measurement. The distance L may be greater than the beam angle γ . Here, the distances $L_{sub.1}$ and $L_{sub.2}$ may be determined as described above regarding $l_{sub.1}$ and $l_{sub.2}$. Using the Pythagorean Theorem, the program **223** may determine the distance L . Additionally, as the beam angle γ from the projection apparatus is known, and the distance $L_{sub.1}$ may be determined as described herein, the distance $L_{sub.3}$ between the center point of the projection beam and the edge of the beam may be determined. For example, using the equation $\tan \gamma = L_{sub.3} / L_{sub.1}$ (wherein γ is $\frac{1}{2}$ the beam angle), the program may determine distance $L_{sub.3}$. Since the user may not be standing at the end **105** of the surface **100**, and the beam of projection is limited, the image of the tape measure may start at point U , the unit of which may be equal to the distance L less distance $L_{sub.3}$. For example, if the user is standing 20 inches from the end of the surface **105**, and the beam projects an image 8 inches across, the point P will show the beginning of the tape measure starting at 16 inches.

[0066] The process may repeat steps **610a** through **610e** in a continuous loop to ensure that the projected image of the tape measure is consistently accurate. In one embodiment, multiple differential calculations may be made and accumulated to provide increased precision to the overall measurement results to the user.

[0067] Optionally, in one embodiment, the process moves to step **612**, where the user may be able to lock the projected image. For example, the user may press the button **215A** or **215A'** (e.g., two quick presses of the button) to lock the image in place. Alternately, the user may be able to effectuate a movement by his head to lock the image (e.g., moving his head in a quick up-and-down movement). By locking the image in place, the user may cause the process to stall at step **620** until the image is unlocked. The user may unlock the image at step **614** by pressing the button, shaking his head, or other means. Once the image is unlocked, the process may return to step **610** as described above.

[0068] The process ends at step **616** with the user deactivating the projection apparatus **200**.

[0069] In another embodiment, a projection apparatus is configured to project an image onto a surface. The image may be based on one or more direct and/or indirect measurements. In other words, one or more direct and/or indirect measurements may be utilized to provide a unique projected perspective image. In one embodiment, the projection apparatus is substantially similar to the projection apparatus **200**, and includes a processor in data communication with a networking device, at least one input/output device, and computer memory that has programming that, when effected by a processor, performs various steps for providing an image. The steps include determining a distance between the projection apparatus **200** and at least one surface, and

projecting an image that is based on the determined distance. The process may be iterative, and it should be understood that the image which may be projected can be dynamic, or may change over time. The image may change incrementally, or may change substantially over time.

[0070] According to another embodiment of the invention, illustrated in FIG. 10, a display apparatus **1000** is disposed near an array receiving surface **1005**, such as a windshield. The array receiving surface **1005** has a first edge **1007** and a second edge **1010**. The display apparatus **1000** has a viewing angle λ (represented by the broken lines in FIG. 10). The display apparatus **1000** may be similar to the projection apparatus **200**, except as is shown and described herein. Here, display apparatus **1000** is operable to determine a distance to a surface **1020** in front of the apparatus **1000**, and to display an image onto the array receiving surface **1005** that is based on information about the surface **1020**. In an embodiment, the information about the surface **1020** is one or more utility lines **1025** which may be disposed above ground or underground. The information may optionally also include the surface **1020** itself. In other words, the image displayed on the array receiving surface **1005** may be of the environmental object **1025**, and/or of the surface **1020**. Accordingly, a user may be presented with an alternative view of the surface **1020** and the environmental objects **1025**. Preferably, however, a user has an unobstructed view the surface **1020** together with a layered image of the environmental objects **1025** in order to provide the user with a useful, real-time accurate image of objects within the field of view of the apparatus **1000**.

[0071] Various methods may be used to determine the information about the surface **1020**. In embodiments, the display apparatus **1000** may be equipped with a camera which may scan the surface **1020** for information which is readily seen by the camera. In other embodiments, the display apparatus **1000** may be equipped with sonar technology, which may be used to send sound waves underground to determine the presence of underground utility lines. Other methods of determining underground objects, whether now known or later developed, may be used to detect the presence of objects on or under the surface **1020**.

[0072] In an embodiment, the surface **1020** may be, for example, human skin (which may also include the muscles beneath the skin). The apparatus **1000** may be able to detect abnormalities under the skin which would otherwise be invisible to the human eye. For example, a doctor may use the apparatus **1000** to scan a person's arm where a broken bone is suspected. An image of the person's bones can be displayed upon the surface **1020** to give the physician an indication of where the breaks are located, and what type of reconstruction is necessary. Utilizing the image, the physician may be able to conduct a closed reduction of the broken bone(s) without requiring more invasive imaging techniques. Additionally, the displayed image can be continuously updated in real time (or substantially in real time) to provide the physician with an up-to-date image of the concerned bone structure.

[0073] FIG. 11 is a flowchart illustrating a closed loop system having various steps performed by the display apparatus **1000** in conjunction with programming effected by a processor. The process starts at step **S1100**, where the display apparatus **1000** detects the presence of a surface **1020**. The surface **1020** may be, for example, an area of earth or an object within the viewing panel of the display apparatus **1000**. At step **S1110**, the apparatus **1000** marks an edge **1022** of the surface **1020**. The edge **1022** is within the viewing panel of the display apparatus **1000**. The apparatus **1000** then determines and marks one or more locations a-n on the surface **1020** at step **S1112**. Optionally, at step **S1114**, the apparatus **1000** may determine the presence of an environmental object (e.g., a utility line **1025**) on or under the surface **1020**. At step **S1116**, the programming determines whether the system has an updated array. If not (e.g., the first time through the process), the apparatus **1000** displays an initial array onto the image receiving surface **1005** at step **S1118**. If the system has an updated array, then the apparatus **1000** replaces the initial array with the updated array at step **S1120**, and the process moves to step **S1122**.

[0074] At step **S1122**, one or more distances **D1.sub.a-D1.sub.n** between the apparatus **1000** and

the locations a-n on the surface **1020** are determined. At step **S1124**, a distance **D2** between the apparatus **1000** and the image receiving surface **1005** is determined. At step **S1126**, a distance **D2** is determined between the apparatus **1000** and the edge **1022** of the surface **1020**. At step **S1128**, the system determines whether a distance **D4** has been previously determined. If not (e.g., the first time through the process) then the process moves to step **S1130**, wherein a distance **D4** between the edges **1005** and **1007** of the image receiving surface **1005** is determined. If the distance **D4** has been previously determined, then the process moves to step **S1132**, where the initial array displayed at step **S1118** is calibrated based on the determined distances **D1**, **D2**, **D3**, and **D4**. Step **S1132** may be similar to step **S610** described above, wherein the processor determines a factor to alter the image based on the distances **D1**, **D2**, **D3**, and **D4**. Finally, at step **S1134**, the display apparatus **1000** displays an updated array on the image receiving surface **1005**. The process then repeats until the apparatus **1000** is deactivated.

[0075] Referring now to FIG. **12**, according to still another embodiment, a light emitting apparatus **2000** is provided. The light emitting apparatus **2000** is similar to the projection apparatus **200**, except as described herein or as may be inherent. Here, the light emitting apparatus **2000** includes a processor **2050** in data communication with a networking device **2100**, one or more input/output devices **2150**, and a memory **2200**. The memory **2200** includes programming **2250** operable to achieve the functionality of the light emitting apparatus **2000** as described herein. The networking device **2100** can communicate over a network **2500** with a remote sharing apparatus **2700** (which may be multiple receiving panels **2700**), which likewise has a networking device **2850**, a processor **2750**, and an input/output device **2800**. As is described in greater detail below, the sharing apparatus **2700** may receive information from the light emitting apparatus **2000** and sharing the information with a third party.

[0076] According to an embodiment, the input/output device **2150** includes a light source **2150a** operable to output light waves. It shall be understood that light waves as used herein includes electromagnetic radiation of any wavelength, whether the electromagnetic radiation is in the visible spectrum or not. The light source may include at least one lens for directing the light waves toward a receiving panel. As used herein, the term “panel” is not limited to a flat or level surface, but may include a flat or level surface, as well as a surface exhibiting undulations or waves. Therefore, a receiving panel can be any surface towards which the light waves are directed, regardless of the surface characteristics exhibited by the surface.

[0077] As shown in FIGS. **13A** and **13B**, the light source **2150a** emits light waves **2152** toward a receiving panel **2160** at a projection angle Ω . The receiving panel **2160** may be transparent or opaque. The light waves **2152** are received upon the receiving panel **2160**. In embodiments where the light waves **2152** are in the visible spectrum, the light waves **2152** illuminate a pattern upon the receiving panel **2160** which may be visible to the naked eye. In some embodiments, the light waves **2152** may not be in the visible spectrum, and therefore the light waves **2152** received by the receiving panel **2160** may not be visible to the naked eye. The light waves **2152** may be adjusted to provide a focused pattern as described in greater detail below regarding FIG. **15**.

[0078] In one embodiment, illustrated in FIGS. **14A** and **14B**, at least two light sources **2150a** and **2150b** are incorporated into the light emitting apparatus **2000**. Here, the light sources **2150a** and **2150b** emit two different light waves **2152a** and **2152b** towards the receiving panel **2160**. The first light wave **2152a** hits the receiving panel **2160** at point P. The second light wave **2152b** hits the receiving panel at point P2. The respective light waves **2152a** and **2152b** may then be adjusted such that the light waves **2152a** and **2152b** meet at a focal point FP at the receiving panel **2160**. In embodiments, the meeting of the light waves **2152a** and **2152b** at the focal point FP results in a haptic stimulation at the receiving panel **2160**.

[0079] The input/output device **2150** may optionally additionally include a camera **2152c** operable to capture images or video of a display. The images or video from the camera **2152c** may be transmitted (e.g., over the network **2500**) to the sharing apparatus **2700**. The sharing apparatus

2700 may include a projector (e.g., as the output device **2800**) for projecting the image from the camera **2152c** to a third party. In embodiments, the camera **2152** transmits, and the sharing apparatus **2700** receives and projects, the images in real time to the third party. In another embodiment, the images and/or video from the camera **2152** are stored (e.g., in memory **2200**) for later use.

[0080] FIG. **15** illustrates a closed loop system having various steps performed by the light emitting apparatus **2000** in conjunction with the programming **2250** effected by the processor **2050**. The process starts at step **S3100**, where the light emitting apparatus **2000** activates a light source **2150** to emit a light wave **2152** in the direction of a receiving panel **2160**. Optionally at step **S3100**, a camera **2150** may also be activated to begin filming the receiving panel **2160**. At step **S3110**, the light emitting apparatus **2000** determines a distance **D1** between the light source **2150** and the receiving panel **2160**. At step **S3112**, the apparatus **2000** determines an edge of the panel **2160**. Then, at step **S3114**, the apparatus **2000** determines the distance **D2** from the edge of the panel **2160** and the point **P** where the light wave **2152** hits the receiving panel **2160**. At step **S3116**, the programming determines the preferred point **DP** at which the light wave **2152** should hit the panel **2160**, and the distance **D3** from the edge of the panel **2160** to the preferred point **DP**. The preferred point **DP** may be based on environmental conditions in the area of the panel **2160**, such as the current lighting, the angle of projection Ω , the distance **D1** from the light source **2150**, etc. Because the environment changes, the preferred point **DP** may also change. Accordingly, the process may be performed iteratively to ensure that the light wave **2152** always hits the panel **2160** at the preferred point **DP**. If **D2** does not equal **D3**, then at step **S3118**, the light source **2150** is adjusted accordingly. At step **S3120**, the light source **2150** is activated to provide an adjusted light wave in the direction of the receiving panel **2160**. As noted above, because the preferred point **DP** may be in constant change due to a change in environmental conditions, the process then returns to step **S3114** to ensure that the light wave **2152** always hits the panel **2160** at the preferred point **DP**. The process may repeat until the apparatus **2000** is deactivated.

[0081] As described above, the apparatus **2000** may include more than one light source **2150a**, **2150b**. FIG. **16** illustrates a closed loop system having various steps performed by the light emitting apparatus **2000** in conjunction with the programming **2250** effected by the processor **2050** wherein the light emitting apparatus includes more than one light source **2150a**, **2150b**. Here, the process begins at step **S4100**, where the light sources **2150a** and **2150b** are concurrently activated. Optionally at step **S4100**, a camera **2150** may also be activated to begin filming the receiving panel **2160**. At step **S4110**, the apparatus **2000** determines a distance **D1** between the light sources **2150a** and the receiving panel **2160**. At step **S4112**, the apparatus **2000** determines the respective edges of the panel **2160**. Then, at step **S4114**, the apparatus **2000** determines the distances **D2a** and **D2b** from the respective edges of the panel **2160** and the points **P** and **P2** where the light wave **2152a** and **2152b** respectively hit the receiving panel **2160**. At step **S4116**, the programming determines the focal point **FP** at which the light waves **2152a** and **2152b** should combine to hit the panel **2160**, and the distances **D3a** and **D3b** from the edges of the panel **2160** to the focal point **DP**. If **D2a** does not equal **D3a**, then at step **S4118**, the light source **2150a** is adjusted accordingly. Likewise, if **D2b** does not equal **D3b**, then the light source **2150b** is adjusted accordingly. At step **S4120**, the light sources **2150a**, **2150b** are activated to provide adjusted light waves **2152a'**, **2152b'** in the direction of the receiving panel **2160**, which converge at focal point **FP**. Because of environmental changes, focal point **FP** may also change. Accordingly, the process may be performed iteratively to ensure that the light waves **2152a'**, **2152b'** always converge at the focal point **FP**. Thus, the process may return to step **S4114** and repeat until the apparatus **2000** is deactivated. It shall be understood that the focal point **FP** may, but need not be directly adjacent the panel **2160**.

[0082] In a further embodiment, illustrated in FIG. **17**, an apparatus **5000** is designed as a scanning and marking apparatus to aid in identifying and optionally reducing irregularities on a surface or in an environment. The apparatus **5000** may be substantially similar to the apparatus **200**, **1000** as

described herein, except as may be shown and/or described. Here, the apparatus **5000** is configured as a scanning device equipped with programming for recognizing abnormalities. Consider, for example, a hood **5010** on a vehicle, as illustrated in FIG. 17. The hood **5010** may have one or more irregularities **5012**. The irregularities **5012** may be dents caused by hail damage (or any other impact), areas where the paint is chipped or discolored, or any other irregularity. The irregularities **5012** may be significant such that they can be seen by the human eye. However, in embodiments, the irregularities **5012** may be difficult to see. Accordingly, the scanning apparatus **5000** may be utilized to scan an area to highlight irregularities, and particularly those irregularities that may be difficult to see.

[0083] In an embodiment, the scanning apparatus **5000** may project an image onto the surface **2010**. The image may be visible or invisible, and in embodiments may comprise an array of visible or invisible light (e.g., in the ultraviolet range). In another embodiment, the scanning apparatus **5000** may instead project sound waves toward the surface **5010**. In still other embodiments, other forms of waves (e.g., thermal, such as thermographic signal reconstruction) to identify irregularities on the surface **5010**. The scanning apparatus **5000** may include a built-in camera or other image scanner/reader, or the image scanner/reader may be separate from the scanning apparatus **5000**. The image scanner is configured to review the image projected onto the surface **5010**, or to review reflected waves bouncing off the surface **5010**, as the case may be. At the areas of irregularities **5012**, the image (or reflective waves) may be distorted as seen by the image scanner. Programming associated with the apparatus **5000** may identify the areas of irregularities based on image from the image scanner. Optionally, the scanning apparatus **5000** may mark the surface **5010** at the areas of irregularity. For example, a light (e.g., laser, image, etc.) may project from the scanning apparatus **5000** at each of the areas of irregularity to alert a user to the problematic area. In embodiments, the marking of the irregularity may be permanent (e.g., via paint, laser etching, etc.) or semi-permanent (e.g., via chalk, excitation of smart color-changing particles, etc.). The scanning apparatus **5000** may be equipped with means (e.g., a sprayer) for depositing the paint or other marking medium, and may therefore be mobile to allow the scanning apparatus **5000** to adequately move about the area of the surface **5010** so as to mark the irregularities.

[0084] The programming associated with the scanning apparatus **5000** may be tailored for a particular environment; however, the operation of the scanning apparatus **5000** remains virtually unchanged regardless of the environment. For example, in one embodiment, the scanning apparatus **5000** may be able to scan, identify abnormalities, and optionally mark abnormalities on roof shingles (e.g., due to hail damage). In another embodiment, the scanning apparatus **5000** is programmed to scan and find abnormalities on a road surface (e.g., potholes). Here, the scanning apparatus **5000** may be configured to direct light to a pothole so as to alert the driver to avoid hitting the pothole. In certain embodiments, the scanning apparatus **5000** may be in communication (e.g., wired or wireless communication) with one or more automotive systems which may be able to proactively adjust in advance of the pothole (e.g., increase the damping capabilities of the shock absorbers) so that the vehicle is not harmed, or the harm can be minimized.

[0085] In still another embodiment, the scanning apparatus **5000** may be configured to identify abnormalities on a person's skin (e.g., the presence of concerning moles) or in a person's mouth (e.g., movement of gums as an indication of gingivitis). Here, the programming may utilize self-learning in order to more accurately identify abnormalities. For example, the programming may initially establish a baseline scan for a person. When the scanning apparatus **5000** conducts subsequent scans, the subsequent scans may be compared against the baseline to determine the presence of previously undetected abnormalities, or areas where previously identified abnormalities have changed since the last scan. The scanning apparatus **5000** may mark the areas of new abnormalities and/or areas where abnormalities have changed such that a physician can very quickly and easily identify areas of interest for further review. The scans can be stored in a secure database which may be accessible by a patient and/or the patient's healthcare providers.

[0086] In yet another embodiment, the scanning apparatus **5000** may be configured to recognize unsafe conditions. For example, the scanning apparatus **5000** may be utilized in a warehouse, and can identify where packages are stacked incorrectly (e.g., via identifying and/or measuring packages as described below, by determining that an object protrudes outside of a predetermined threshold area, etc.). The scanning apparatus **5000** can mark the area of the unsafe condition (e.g., by projecting an image onto the ground alerting people to stay away from the unsafe area), and can further sent an alert to ensure that the situation is rectified. As shall be understood by those of skill in the art, other dangerous conditions in many different environments can be identified, marked, and expediently rectified.

[0087] In a further embodiment, the scanning apparatus **5000** may utilize night vision (e.g., image intensification, thermal imaging, etc.) to scan an area to identify the presence of objects that would otherwise be imperceptible to the human eye. For example, the scanning apparatus **5000** may utilize night vision to sense and identify an assailant in an attempt to get away from law enforcement. Once the scanning apparatus **5000** has identified the location of the assailant, the scanning apparatus may mark the location of the assailant, including movement of the assailant. In one embodiment, the scanning apparatus **5000** is elevated (e.g., on a tower or helicopter), and a laser is used to mark the assailant such that law enforcement on the ground can quickly and easily locate the assailant. In another embodiment, the marking may be invisible to the human eye, and the location information can be provided to law enforcement officers (e.g., latitude/longitude, street names, etc.) such that the assailant is not made aware of the marking.

[0088] Moving on, according to another embodiment, the scanning apparatus **5000** can aid in building processes. For example, the scanning apparatus **5000** may project an image onto an area where it is desirable to have a level surface. The scanning apparatus **5000** may, via the projected image and image scanner, identify low-lying areas or areas that are not level. The scanning apparatus **5000** may mark the low-lying area such that additional material may be provided. In one specific implementation, the scanning apparatus **5000** may be utilized for road building. The scanning apparatus **5000** can scan, locate areas that are not level, and optionally mark those areas so that the road can be leveled. This may be particularly useful where several layers of material are deposited onto a surface, as one can quickly and easily ensure that each layer is level before moving onto the next.

[0089] In still another embodiment, a system is designed as a scanning and measuring and/or counting apparatus. Here, the scanning apparatus may be substantially similar to the scanning apparatus described herein except as may be shown or described herein. The scanning apparatus may include programming for determining edges of a package and/or measuring the edges of the package. The size of the package can be utilized by the programming to verify, for example, payment for the item, or where the package should be stored. The programming may be able to measure and identify individual packages, as well as identify size and dimensions of a collective group of packages. For example, the scanning apparatus can quickly and easily assess the number of packages for inventory purposes. In embodiments, the scanning apparatus can scan the package prior to, or as, a shopper is placing the item into his or her cart. The scanning apparatus can identify the package, and determine the correct cost of the item. In another embodiment, the scanning apparatus can identify one or more packages after they have been placed in a shopping cart. Inventory can be quickly and easily updated based on the information, and the sale price of the goods can be quickly and easily affirmed.

[0090] In still another embodiment, a scanning and counting system may receive information from a user related to a scanning area. For example, a pharmacist may enter prescription information into the system, such as the number of pills prescribed by the doctor. The pharmacist may then dispense the prescription onto the scanning area. The system may scan the area and determine the edge of the pills. Once the edge of each of the pills has been detected, the system may then project an initial image onto the scanning area. The initial image may include, for example, one or more images near

each of the pills on the scanning area. In one example, the projection is a circle surrounding each of the pills. The system may then scan the scanning area to determine the number of pills atop the scanning area. The system may then access the information into the system to determine the correct number of pills to fill the prescription. The system may then project a second image onto the scanning area. The second image may include for example, one or more images near each of the pills on the scanning area. In one example, the second image is a dot projected upon each of the pills. In embodiments, the projected dots may be configured based on the number of pills in the scanning area. For example, if the prescription calls for 12 pills, then a green dot may be projected upon each of 12 pills. Any excess pills in the scanning area may receive a red dot. Of course, images other than dots may be projected onto or near the pills. If less than 12 pills are in the scanning area, a third image may be projected onto the scanning area to alert the pharmacist that additional pills are needed in the scanning area. The pharmacist may then disperse additional pills onto the scanning area and the process may be repeated until at least as many pills are on the scanning area as required by the prescription. In some embodiments, a means for separating excess pills may be included as part of the system. For example, an arm may be activated to separate the pills with the green dots from the pills with the red dots. The pills with the green dots may then be placed into the container for the patient, while the excess pills are returned to the storage container. [0091] Thus has been described systems, methods, and apparatus for projecting an image onto a surface. Many different arrangements of the described invention are possible without departing from the spirit and scope of the present invention. Embodiments of the present invention are described herein with the intent to be illustrative rather than restrictive. Alternative embodiments will become apparent to those skilled in the art that do not depart from its scope. A skilled artisan may develop alternative means of implementing the disclosed improvements without departing from the scope of the present invention.

[0092] Further, it will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations and are contemplated within the scope of the claims. Not all steps listed in the various figures and description need to be carried out in the specific order described. The description should not be restricted to the specific described embodiments.

Claims

1. A wearable article, comprising: a projection apparatus embodied in a housing, the projection apparatus comprising a processor in data communication with computer memory and a processor, the computer memory comprising a program having machine readable instructions that, when effectuated by the processor, perform the following steps: (a) accessing a data file having an intended display size; (b) determining a position of the projection apparatus relative to a display surface, the position including distance and angle; (c) using the position from step (b) to determine a factor by which data from the data file must be altered for the intended display size to correspond to an actual display size; (d) altering the data from the data file by the factor determined in step (c); (e) causing the projector to project the altered data from step (d) onto the display surface whereby the actual display size corresponds to the intended display size; and (f) repeating steps (b) through (e).
2. The wearable article of claim 1, wherein the display surface is generally planar.
3. The wearable article of claim 1, wherein the wearable article is a hat.
4. The wearable article of claim 1, wherein the wearable article is a pair of goggles or glasses.
5. The wearable article of claim 1, wherein the wearable article is a clip.
6. A projection apparatus for temporarily displaying images on human skin of a person to be projected upon, the projection apparatus comprising: a projector; computer memory; and a processor in data communication with the projector and the computer memory; wherein the

computer memory comprises a program having machine readable instructions that, when effectuated by the processor, perform the following steps: (a) accessing a data file; (b) determining an intended display size based on at least one characteristic of the person to be projected upon; (c) determining a position of the projection apparatus relative to the human skin of the person to be projected upon, the position including distance and angle; (d) using the position from step (c) to determine a factor by which data from the data file must be altered for the intended display size to correspond to an actual display size; (e) altering the data from the data file by the factor determined in step (d); (f) projecting the altered data from step (e) onto the display surface whereby the actual display size corresponds to the intended display size; and (g) repeating at least steps (c) through (f).
